Bang-time perturbations and redshift drift in a parabolic LTB model: a reproducible light-cone calculation

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Abstract

We calculate the response of redshift drift to a radially varying bang time in an exactly solvable, parabolic Lemaître–Tolman–Bondi (LTB) dust spacetime. The observational drift is propagated between neighboring light cones; the familiar Friedmann–Lemaître–Robertson–Walker (FLRW) endpoint formula is not assumed in the inhomogeneous model. We derive the light-cone and linear-response equations, and provide a reproducible integration for two smooth, centrally regular bang-time perturbations. At equal luminosity distance, both sampled responses are negative over baseline redshifts $0.1 \le z \le 1.5$ and show no zero crossing. This is an illustrative calculation, not a fit to cosmological data or a general uniqueness result. A single equal-distance comparison also differs from compensating the complete distance–redshift relation by adjusting the other free LTB function. These qualifications limit the observable inference from the calculation.

1 Scope and relation to earlier work

Redshift drift in spherically symmetric spacetimes and its use with distance data were established by Uzan, Clarkson and Ellis [1]. Yoo, Kai and Nakao developed the LTB drift calculation and studied void models [2]. Exact growing and decaying density modes in LTB dust have been analyzed by Sussman [3]. The inverse problem for LTB observables has additional existence and regularity conditions [4]. We therefore claim neither a new general drift equation nor a unique observational signature. The purpose here is to correct an endpoint-formula error in an earlier unpublished draft, spell out an explicit parameter derivative, and give a small reproducible test case.

In units $c = G = 1$, the LTB metric and evolution equation are

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \frac{R'(t,r)^2}{1+2E(r)}\mathrm{d}r^2 + R(t,r)^2\mathrm{d}\Omega^2, \tag{1}$$

$$\dot{R}^2 = \frac{2M(r)}{R} + 2E(r), \qquad R(t_{\mathrm{B}}(r), r) = 0. \tag{2}$$

Dots and primes denote partial derivatives at fixed r and t, respectively. A radial coordinate choice removes one of M, E, t_{B} as an independent function. A nonconstant t_{B} represents the decaying mode in the conventional LTB dust mode decomposition [3]; its additive constant sets the time origin. Throughout this paper the mass-coordinate gauge and the profiles below are specified explicitly.

2 Drift on neighboring radial light cones

Put $S(r) = \sqrt{1+2E(r)}$, $A = R'/S$, $B = \dot{R}'/S$, and $C = \ddot{R}'/S$. On an incoming radial ray from a central, comoving observer at $r = 0$, the emission time $t(r)$ and observed redshift $z(r)$ satisfy

$$\frac{\mathrm{d}t}{\mathrm{d}r} = -A, \qquad \frac{\mathrm{d}z}{\mathrm{d}r} = (1+z)B, \qquad t(0) = t_0, \quad z(0) = 0. \tag{3}$$

Compare two rays received at times separated by δt_0, from the *same comoving emitter* r. Their time separation $T(r) = \partial t(r; t_0)/\partial t_0$ obeys $T' = -BT$, $T(0) = 1$, hence $T = 1/(1+z)$. Let $Z(r) = \partial z(r; t_0)/\partial t_0$ be drift per unit central proper time. Differentiating Eq. (3) with respect to t_0 at fixed r gives the exact propagation equation

$$Z' = BZ + (1+z)CT = BZ + C, \qquad Z(0) = 0, \tag{4}$$

or, equivalently,

$$\boxed{Z(r) = (1+z(r)) \int_0^r \frac{C(\bar{r})}{1+z(\bar{r})}\,\mathrm{d}\bar{r}.} \tag{5}$$

This result carries the information along the ray. In a homogeneous FLRW limit it reduces to $Z = (1+z)H_0 - H(t_e)$. In a general LTB geometry, replacing Eq. (4) by $(1+z)H_0 - \dot{R}'/R'|_e$ is not justified: the gradient of expansion along the ray contributes. For an isotropic central observer, reciprocity gives

$$D_L(r) = (1+z(r))^2 R(t(r), r). \tag{6}$$

3 Linear response and what "fixed distance" means

Let p denote a model parameter and $\boldsymbol{y} = (t, z, Z)$. With $\boldsymbol{y}' = \boldsymbol{F}(\boldsymbol{y}, r, p)$ from Eqs. (3) and (4), its first-order variation $\boldsymbol{u} = \partial_p \boldsymbol{y}|_r$ is specified, without finite differences, by

$$\boldsymbol{u}' = \underbrace{\begin{pmatrix} -A_t & 0 & 0 \\ (1+z)B_t & B & 0 \\ B_t Z + C_t & 0 & B \end{pmatrix}}_{\boldsymbol{F}_{\boldsymbol{y}}} \boldsymbol{u} + \begin{pmatrix} -A_p \\ (1+z)B_p \\ B_p Z + C_p \end{pmatrix}, \qquad \boldsymbol{u}(0) = \boldsymbol{0}. \tag{7}$$

All partial derivatives in this equation are at fixed t, r; in particular, A_p includes any parameter dependence in S. Equation (7) is the analytic variational equation used to define the linear response; the public numerical example evaluates its derivative by symmetric finite differences and checks step convergence.

At a given emitter coordinate the distance response is

$$\partial_p D_L|_r = (1+z)^2(R_t t_p + R_p) + 2(1+z)Rz_p, \qquad \frac{\mathrm{d}D_L}{\mathrm{d}r} = 2(1+z)z_r R + (1+z)^2(R_t t_r + R_r). \tag{8}$$

Here R_p and R_r are partial derivatives, whereas $\mathrm{d}D_L/\mathrm{d}r$ is a total derivative along the ray. If $\mathrm{d}D_L/\mathrm{d}r \neq 0$, implicit differentiation yields the response at a *single equal luminosity distance*,

$$\boxed{\left.\frac{\partial Z}{\partial p}\right|_{D_L} = Z_p|_r - \frac{Z_r}{\mathrm{d}D_L/\mathrm{d}r}\,\partial_p D_L|_r.} \tag{9}$$

For comparison, $\partial_p Z|_z = Z_p|_r - (Z_r/z_r)z_p|_r$. Neither condition requires $D_L(z)$ to be identical between the two models. Keeping the *whole function* $D_L(z)$ fixed would require a compensating functional perturbation in M or E and is a distinct inverse problem. Equations (7)–(9) also show that a zero-crossing location cannot be inferred from the background profile alone without specifying the allowed perturbation and compensation.

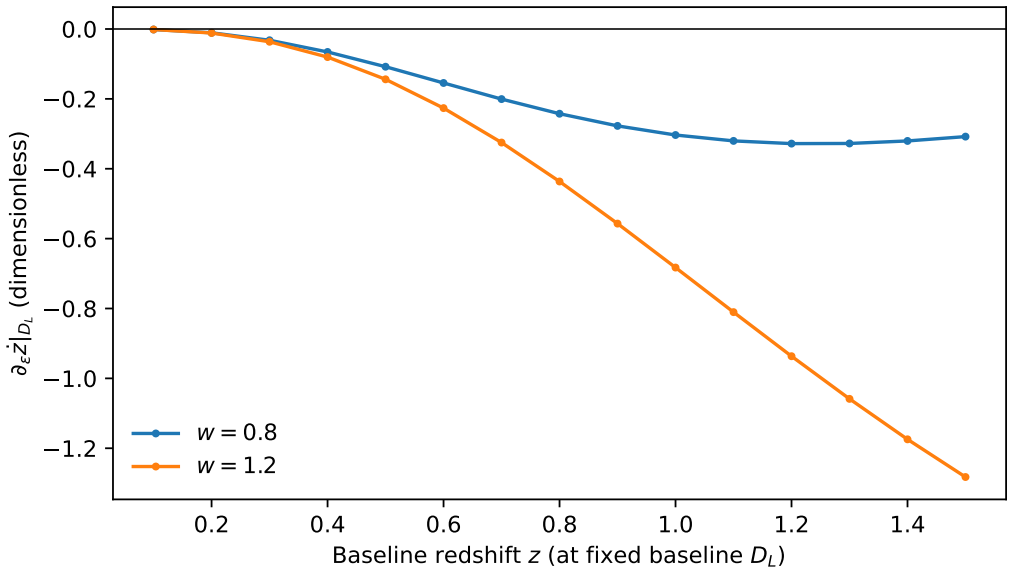


Figure 1: Dimensionless bang-time response at equal D_L for the two specified f_w. The horizontal coordinate labels the *baseline* redshift. Lines guide the eye between 15 sampled points; no observational uncertainties are implied.

4 Exactly specified illustrative model

Set $t_0 = 1$, $E(r) = 0$, $M(r) = 2r^3/9$, and $t_{\rm B}(r) = \epsilon f_w(r)$, with

$$f_w(r) = r^2 \exp[-(r/w)^2], \qquad w \in \{0.8, 1.2\}. \tag{10}$$

All coordinates and drift values in this section are dimensionless. The two choices of w are perturbation shapes on the *same homogeneous background*, not two realistic void profiles. The even behavior at $r = 0$ makes the center smooth. The exact expanding solution, with $\tau = t - \epsilon f_w(r)$, is

$$\begin{aligned} R &= r\tau^{2/3}, \\ R' &= \tau^{2/3} - \frac{2r\epsilon f_w'}{3\tau^{1/3}}, \quad \dot{R}' = \frac{2}{3\tau^{1/3}} + \frac{2r\epsilon f_w'}{9\tau^{4/3}}, \\ \ddot{R}' &= -\frac{2}{9\tau^{4/3}} - \frac{8r\epsilon f_w'}{27\tau^{7/3}}. \end{aligned} \tag{11}$$

The expansion branch requires $\tau > 0$; the code checks $R' > 0$ and monotonic z, D_L on every sampled light cone. At $\epsilon = 0$, the model is Einstein–de Sitter (EdS) and has the exact check

$$Z_{\rm EdS}(z) = \frac{2}{3}[(1+z) - (1+z)^{3/2}]. \tag{12}$$

At $\epsilon \neq 0$ we hold M, E, t_0 fixed, integrate to $z = 1.6$, and compare drift values at the *baseline* $D_L(z)$ for $z = 0.1, 0.2, \ldots, 1.5$. The derivative uses $[Z(+h) - Z(-h)]/(2h)$ at $h = 0.002, 0.001, 0.0005$. The plotted result uses $h = 0.0005$.

The maximum absolute baseline error against Eq. (12) on this grid is 4.4×10^{-13}. An independent two-observer-time finite difference of redshift at fixed $r = 0.8$, with $\epsilon = 0.001$ and time step 10^{-5}, agrees with the propagated drift to 3.3×10^{-11}. For $\epsilon = 0.001$, the first-order maximum fractional distance change at fixed baseline z is approximately 2.20×10^{-4} ($w = 0.8$) or 3.78×10^{-4} ($w = 1.2$). Thus

Table 1: Diagnostics produced by `python analysis.py`. Extremes and convergence refer to the 15 sampled distances, not a continuum bound.

w	$\min \partial_\epsilon Z\|_{D_L}$	$\max \partial_\epsilon Z\|_{D_L}$	max. change on halving h
0.8	-0.328043	-0.001464	1.39×10^{-7}
1.2	-1.282333	-0.001490	1.81×10^{-7}

the equal-distance operation is numerically distinct from holding the complete distance relation fixed. Table 1 and Fig. 1 support the narrow statement that these two examples have no sign change at the sampled distances. They do not establish a sign theorem for arbitrary f, nonparabolic backgrounds, or compensated models.

5 Interpretation and limitations

The present parabolic EdS background has no growing-mode void, dark energy, realistic expansion calibration, observational covariance, or early-Universe consistency test. Bang-time variations can generate strong early-time effects and are constrained by other observations; this manuscript does not estimate allowed amplitudes. The plots give derivatives in chosen dimensionless coordinates, not a predicted velocity drift in cm/s over a survey baseline. A physical forecast needs a calibrated model and nuisance effects, along with a functional distance compensation and a treatment of observational uncertainties. This preprint therefore provides a corrected calculational baseline, not evidence for a detected or uniquely isolatable decaying mode.

Reproducibility

The accompanying `analysis.py`, dependency list, figure and machine-readable numbers fix every numerical input. Execute `python analysis.py` followed by `pdflatex paper.tex` twice. No observed data or generated artwork enters the result.

References

[1] J.-P. Uzan, C. Clarkson and G. F. R. Ellis, "Time Drift of Cosmological Redshifts as a Test of the Copernican Principle," *Phys. Rev. Lett.* **100**, 191303 (2008), doi:10.1103/PhysRevLett.100.191303.

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[3] R. A. Sussman, "Invariant characterization of the growing and decaying density modes in LTB dust models," arXiv:1305.3683 (2013).

[4] C.-M. Yoo, "A Note on the Inverse Problem with LTB Universes," *Prog. Theor. Phys.* **124**, 645–667 (2010), arXiv:1010.0530.